

Amogh Gulati

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Education

NYU Courant Institute of Mathematical Sciences

New York, NY | Sep 2025 – May 2027

M.S. Computer Science

GPA: 3.8/4.0

IIIT Delhi

New Delhi, India | Aug 2017 – May 2021

B.Tech. Computer Science and Engineering

GPA: 8.71/10

Skills

Languages: Python, C++, SQL, Bash

ML & Data: PyTorch (DDP), ONNX, Weights & Biases, Xarray, Zarr, Pandas, NumPy

Infrastructure: Slurm, Docker, AWS (Lambda, SageMaker, S3, SQS, CloudFront), Elasticsearch, Kibana

Experience

M²LInES, NYU Courant | *Graduate Research Assistant*

New York, NY | Oct 2025 – Present

- Improved [Samudra](#), an ML ocean emulator: skip-connection dropout reduced rollout RMSE by up to **15%** on velocity and **8.5%** on sea surface height; ConvNeXt-axial attention blocks reduced RMSE across all variables and depths.
- Developing replay-buffer training and spectral losses to address spatial smoothing and long-term trend errors in 100-year ocean rollouts.

Gan.ai (Sequoia-backed) | *Founding ML Engineer*

Noida, India | Jan 2021 – Oct 2024

- Built and owned the company's ML stack from scratch, spanning TTS model training, production inference, and deployment; led the ML infrastructure team.
- Trained and deployed TTS models (NaturalSpeech2, FastSpeech2, Tortoise) for English and 6 Indic languages, serving **200k+ daily inferences**.
- Designed AWS deployment pipelines using Lambda, SageMaker, Docker, and ONNX, reducing cold-start latency from **60s to 4s**.
- Owned delivery of personalized video ads for [Samsung](#) and AI-generated birthday songs for [Cadbury](#).
- Established security and operational processes and compiled audit evidence for the company's **SOC 2 Type II** compliance.
- Co-inventor on **2 pending US patent applications** for AI voice-over and multimedia generation.

Precog Research Group, IIIT Delhi | *Undergraduate Researcher*

New Delhi, India | Jun 2019 – Jan 2021

- Built ELK pipelines to automate collection of **100M+ tweets** for social media research.
- Analyzed emotions and misinformation in COVID-19 and political discussions on social media, resulting in publication (below).

Qualcomm | *Software Engineering Intern*

Remote | May 2020 – Aug 2020

- Extended debugging features in the Hexagon IDE and automated build-goal validation with Python.

Publications & Patents

[Google Scholar](#)

- Varadhan, P. S., **Gulati, A.**, Sankar, A., Anand, S., et al. (2025). [Rethinking MUSHRA: Addressing Modern Challenges in Text-to-Speech Evaluation](#). *TMLR*.
- Aggrawal, P., Jolly, B. L. K., **Gulati, A.**, et al. (2021). [Psychometric Analysis and Coupling of Emotions Between State Bulletins and Twitter in India During COVID-19 Infodemic](#). *Frontiers in Communication*.
- Prabhu, A., Guhathakurta, D., Subramanian, M., Reddy, M., Sehgal, S., Karandikar, T., **Gulati, A.**, et al. (2021). [Capitol \(Pat\)riots: A Comparative Study of Twitter and Parler](#). *arXiv:2101.06914*.
- Gulati, A.**, Joshi, B., Jain, C., and Shukla, J. (2020). It's Not What They Play, It's What You Hear: Understanding Perceived vs. Induced Emotions in Hindustani Classical Music. *ICMI '20 Companion*.
- Gulati, A.**, Siddhartha, S., Barman, M. P., Bhatia, A., and Bhooshan, S. (2025). [Voiced-over Multimedia Track Generation](#). *US Patent App. 18/842,687* (pending).
- Siddhartha, S., Bhatia, A., **Gulati, A.**, Barman, M. P., and Bhooshan, S. (2025). [Seamless Multimedia Integration](#). *US Patent App. 19/242,615* (pending).